Lambda-Flight: Coherence Transport as a Measured Residual Closure in Driven Open Physical Systems

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Abstract

Defines Lambda-Flight as a disciplined framework for testing coherent transport in driven open physical systems. A scalar field Lambda = Lambda0 ln C is built from measurable coherence factors, but it is not treated as a new force: the transport term chi grad Lambda is allowed only as a residual closure after accepted baseline dynamics, single-observable models, and equal-feature latent baselines fail out of sample. The framework gives operational definitions, a drift-diffusion/relaxation closure, the flight numbers Pe_Lambda and St_Lambda, a nested baseline hierarchy B0-B4, negative controls, an anti-precursor-hunting rule, and six test cards each carrying an

> *explicit kill clause. A deterministic toy pilot is included only as a numerical calibration of the Pe/St regime map; it is not evidence for the framework in nature.*

> ***Status note:*** *working preprint (v1.0). No Zenodo record yet; the DOI will be added when the record is published. The framework should be treated as pre-registered methodology until at least one platform test has been run against a real accepted baseline.*

1. Scope

FRC 787.787 is a framework paper for coherent transport in driven open physical systems. It is **not** a claim of a new fundamental force, a replacement for accepted equations, or a license to reinterpret famous experiments after the fact.

The framework asks one testable question: does a measured coherence field improve held-out prediction after the accepted baseline has already been given a fair chance? If no, Lambda-Flight fails for that platform. If yes, the result is a nested residual law, not a discovery that ordinary dynamics were absent.

This is the consistency lock with the 100-series. [[FRC-100-003]] and [[FRC-100-006]] place outcome statistics and collapse-like behavior at the level of open-system basin dynamics; [[FRC-100-007]] audits the Lambda field as a coherence geometry; [[FRC-566-030]] treats entropy–coherence reciprocity as an open current rather than a closed conservation identity. 787.787 therefore uses Lambda as a *measured* coherence layer: computed from observables, compared against baselines, and killed when it adds no predictive information.

2. Definitions and Coherence Factors

On a domain $\Omega\subset\mathbb{R}^d$, the minimal measured fields are a density $\rho(x,t)$, an order parameter $\psi(x,t)$, a drive $E(t)$, and a positive dimensionless coherence functional $C(x,t)>0$. The scalar field is

$$\Lambda(x,t)=\Lambda_0\ln C(x,t), \qquad C(x,t)>0,$$

with a useful multiplicative construction

$$C=\prod_{j=1}^{m} C_j^{w_j}, \qquad {\Lambda\over \Lambda_0}=\sum_{j=1}^{m} w_j\ln C_j .$$

Admissible factors C_j include amplitude regularity, local phase order, spectral concentration, defect suppression, correlation persistence, dissipation regularity, and boundary matching — all platform-specific and measurable. The weights and normalizations **must be fixed before testing** or fitted only on training data. A Lambda field built after seeing the event is a visualization, not evidence.

3. Operational Lambda-Flight

For a coherent structure with measured intensity $q(x,t)$, center $X(t)$, and radius $R(t)$, and an internal coherence score $K(t)$, the structure is a Lambda-Flight candidate over $[0,T]$ only if:

1. $K(t)$ stays above a preselected threshold for most of $[0,T]$;
2. $R(t)$ grows more slowly than the matched diffusive or dissipative baseline;
3. the residual velocity after known baseline forces are removed is predicted by the precomputed $\nabla\Lambda$ field **on held-out data**.

The third condition is the load-bearing one. Persistent motion alone is not Lambda-Flight.

4. Minimal Closure

The minimal continuum closure is a nested extension of ordinary transport:

$$\partial_t\rho+\nabla\cdot(\rho u) = \nabla\cdot\left(D_\rho\nabla\rho-\chi\rho\nabla\Lambda\right)+S_\rho .$$

The term $\chi\rho\nabla\Lambda$ is a **residual closure**, permitted only when the accepted baseline transport model has been specified, its residuals measured, $\nabla\Lambda$ predicts those residuals out of sample, and simpler equal-access alternatives do not explain the same gain. Otherwise $\chi=0$ and the model reduces to the baseline. The Lambda field itself is allowed to lag:

$$\tau_\Lambda(\partial_t\Lambda+u\cdot\nabla\Lambda) = \ell_\Lambda^2\nabla^2\Lambda-\Lambda+\Lambda_{\rm eq}[\rho,\psi,E]+\xi_\Lambda .$$

A linear stability analysis of the two-field model yields the candidate instability threshold $\chi a\rho_0>D(1+\ell^2k^2)$ — a physical claim only after a, χ, and the coherence channel are estimated without test leakage. The variational functional in the paper is a Lyapunov-compatible template under stated assumptions, a discipline check — not a theorem of monotonic free-energy decay.

5. Flight Numbers

With structure size L, Lambda contrast $\Delta\Lambda$, effective diffusivity D, and structure speed V:

\[Pe_\Lambda={v_\Lambda L\over D}\sim{\chi\Delta\Lambda\over D}, \qquad St_\Lambda={\tau_\Lambda V\over L}.\]

Regime	Expected behavior	Kill condition
$Pe_\Lambda<1$	diffusion or baseline transport dominates	claimed flight persists with shuffled or zero Lambda
$Pe_\Lambda>1$, $St_\Lambda\ll1$	adiabatic coherence-guided transport possible	no held-out residual prediction
$Pe_\Lambda>1$, $St_\Lambda\sim1$	lag, overshoot, hysteresis	hysteresis already explained by baseline
$St_\Lambda\gg1$	Lambda decouples from moving structure	no degradation relative to low-St_Λ case

All estimates (χ, $\Delta\Lambda$, D, τ_Λ, V, L) must come from training data or independent calibration.

6. Nested Baselines

Lambda-Flight is evaluated as a nested residual model:

Level	Model	Purpose
B0	persistence baseline	recent trajectory continues
B1	accepted platform equation	PDE, TDDFT, phase-field, Kuramoto, reaction-diffusion, or other domain baseline
B2	best single observable	amplitude, phase order, defect density, energy, or one coherence factor
B3	equal-feature non-Lambda latent model	same observables but no Lambda geometry or relaxation law
B4	Lambda model	$B1+\Lambda,\nabla\Lambda,H_\Lambda,\tau_\Lambda$

Support requires B4 to beat B0–B3 on held-out data by a preselected metric with uncertainty estimates. A visible Lambda map is not support. A post-hoc fit is not support. A famous experiment already explained by B1 is not support.

7. Negative Controls

Every serious test includes: shuffled Lambda labels across trials; time-reversed Lambda fields; spatially shifted Lambda fields; single-factor ablations; equal-feature latent baselines; training/test splits by run, parameter regime, or time block; and no-refit prediction horizons fixed before scoring. If the Lambda model survives only when these controls are omitted, it is not a pillar result.

8. Six Test Cards

Each prediction carries a null and an explicit kill clause:

Prediction	Test	Null	Kill
P1 coherence-gradient drift	residual $\dot X_{\rm res}$ aligns with $\nabla\Lambda$	residuals are baseline noise	no held-out alignment after controls
P2 basin residence	$T_{\rm res}$ scales with $\Delta\Lambda$ in a rare-event regime	residence explained by energy or amplitude	single observable beats Lambda
P3 transition early warning	Lambda curvature, variance, or gap degrades before collapse	baseline observable warns equally early	no lead time on held-out transitions
P4 finite-relaxation hysteresis	lag grows with $\tau_\Lambda r$ under sweeps	standard response delay explains loop	no Lambda-specific scaling
P5 coherence-over-amplitude selection	higher Lambda structure persists when amplitude is matched	amplitude or energy selects structure	amplitude-only model wins
P6 Pe/St collapse	flight fails outside registered Pe/St window	failure is numerical or baseline instability	Pe/St does not stratify outcomes

9. Anti-Precursor-Hunting Rule

787.787 must not be used to claim that a new external result confirms FRC because it contains a delay, a coherent packet, a transition, or a transport anomaly. That is precursor hunting. The rule: (1) Lambda observables defined before scoring the external result; (2) the accepted baseline implemented or quoted as the primary null; (3) the Lambda model makes a separable prediction the baseline does not already determine; (4) the prediction is scored out of sample.

The attosecond-STM gate is the example. A measured tunneling delay is not automatically τ_Λ. If TDDFT already contains the dependence, there is no FRC paper. If a separable Lambda relaxation law beats TDDFT out of sample, that can become a separate confirmation paper.

10. Numerical Pilot: Calibration, Not Evidence

The release includes a deterministic toy pilot (`pilot/pilot_lambda_flight.py`): a conserved one-dimensional packet trying to follow a moving coherence basin, swept over $Pe_\Lambda\in\{0.35,1.20,3.00\}$ and $St_\Lambda\in\{0.10,0.60,1.50\}$. The registered success window is $Pe_\Lambda\ge1$ and $St_\Lambda\le0.6$.

Metric	Best registered case	Worst case
Pe_Λ	3.00	0.35
St_Λ	0.10	1.50
median lag / L	0.448	0.499
final basin mass	0.594	0.450
final width ratio	1.388	1.709
tracking score	0.350	0.205

The pilot is only a calibration of the equations and discretization — **not empirical evidence**. Higher Pe_Λ improves capture, larger St_Λ weakens tracking, and low Pe_Λ cannot be rescued by plotting Lambda after the fact. The separation is modest, and the paper does not oversell it.

11. Conclusion

Lambda-Flight is a narrow claim about residual coherent transport. The framework is useful only when the computed Lambda field improves held-out prediction beyond accepted baselines and simpler alternatives. That restriction is the point: if Lambda adds nothing, the platform kills the claim cheaply. If Lambda survives the nested tests, 787.787 becomes a disciplined bridge between the 100-series coherence program and concrete non-equilibrium physical systems.

Connections

- [[FRC-100-003]] — Collapse as open-system basin dynamics; the outcome-statistics level Lambda-Flight must respect
- [[FRC-100-006]] — Born statistics at the microstate-distribution level, not the rho-level drift
- [[FRC-100-007]] — The Lambda-field audit that warns against using nonlinear drift as a Born-statistics carrier
- [[FRC-100-008]] — Executable Lambda engine; oscillator locking time meets the 787 coherence-lifetime criterion
- [[FRC-566-030]] — Entropy–coherence reciprocity as an open current, the thermodynamic frame for the residual closure

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